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Dear Editors,

Re: Submission of "Stage-conditional shift normalization for subject-robust EEG sleep staging"

We submit the enclosed manuscript for consideration as a research article in *Neurocomputing*.

Sleep stage classifiers lose accuracy when they are transferred to a new patient or clinical center. The common label-free response is to replace the batch-normalization statistics of the target recording. That operation implicitly treats subject shift as one recording-wide offset. Our diagnostic results show that this view is incomplete: after marginal mean correction, a subject probe refitted inside each sleep stage still performs well above chance, whereas the ordinary overall probe is dominated by the removable offset. The finding locates a stage-conditional component of subject dependence that global normalization cannot address.

We introduce stage-conditional shift normalization (SCSN). SCSN estimates a separate first-moment shift for each sleep stage from the model's own soft posterior and subtracts the posterior-weighted shift before classification. It is closed form, requires no target labels, optimizer, or weight update, and contains the conventional marginal correction as one endpoint of a one-parameter family. The method applies unchanged to batch- and group-normalized stagers and leaves the backbone and classifier frozen.

The study evaluates SCSN on Sleep-EDF and the independent HMC Sleep Staging Database, using two backbones and two normalization architectures. The key results are:

1. SCSN reduces the within-stage subject probe from 0.74 to 0.47 on Sleep-EDF EEGNet and from 0.53 to 0.21 on HMC EEGNet;
2. accuracy is preserved on Sleep-EDF and improves by 1.4–1.9 points on HMC;
3. on group-normalized models, SCSN improves accuracy significantly in three of four settings and worst-patient conformal coverage in all four;
4. matched-statistic, mismatched-statistic, hard-label, low-rank, and classifier-null-space controls isolate the mechanism;
5. cross-cohort transfer identifies the practical boundary of the method: a weak target posterior reduces SCSN to marginal correction.

The manuscript reports these qualifications rather than presenting SCSN as universally superior. Code, evaluation scripts, and released result summaries will be available at `https://github.com/YuhaoPeng1103/SleepSCSN`.

The manuscript is original, is not under consideration elsewhere, and has not been published previously. Both authors approved the submission. The authors declare no competing interests, received no specific funding, suggest no reviewers, and request the subscription publication route.

Thank you for considering the manuscript.

Sincerely,

Yuhao Peng
On behalf of Zhiwen Song and Yuhao Peng